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Uni. Roll No.

Program: **B.Tech.**

Semester: **3**

Name of Subject: **Data Structures**

Subject Code: **PCIT-101**

Paper ID: **16040**

14-01-2022(M)

Time Allowed: 02 Hours

Max. Marks: 60

NOTE:

- 1) Each question is of 10 marks.
- 2) Attempt any six questions out of nine
- 3) Any missing data may be assumed appropriately

1. How the performance of a program is directly related to the time space trade-off in algorithms? Discuss the complexity of various search and sorting algorithms in terms of Big O notation.

2. Write down the recursive algorithm for binary search. Analyze and Compare the number of comparisons required to search each element using linear search and binary search in the following list of elements:

45, 39, 8, 54, 77, 38, 24, 16, 4, 7, 9, 20

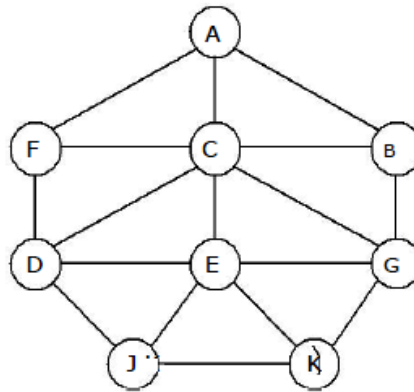
3. Compare and contrast stacks with queues with proper example. What is the importance of representation of an algebraic expression into three different notations? Convert the following algebraic expression into postfix notation using stack: (demonstrate each step carefully)

$a + b * c + (d * e + f) * g$

4. Compare and Contrast circular queue, dequeue and priority queue with help of appropriate examples.

5. How linked list is different from array? Write down the algorithms for various operations that can be performed on circular linked list. .

6. Discuss various methods of representation of graphs with appropriate demonstrations and graphical notations.
7. Traverse the graph shown below in breadth first order and depth first order:



8. Form a heap from the set of elements (40, 80, 35, 90, 45, 50, 70) and sort the data using heap sort algorithm.
9. Briefly explain hashing. What are the properties of a good has function? Explain different hashing techniques with help of appropriate examples.
